



Data Engineering

SETUP PYTHON



What is Python?

 Python is a programming language that lets humans talk to computers in a way that's so simple, even your future self won't hate you for writing it. 😊

How to Install Python on Windows

- Go to the official Python website's [Windows downloads page](#) and download one of the Stable Releases of Python.



The screenshot shows the Python website's navigation bar with the Python logo and the word "python" in white on a dark blue background. Below the logo are three buttons: "About", "Downloads", and "Documentation". The "Downloads" button is highlighted. Below the navigation bar, the breadcrumb "Python >>> Downloads >>> Windows" is visible. The main heading is "Python Releases for Windows". Below this, there is a list of releases. The first release is "Latest Python 3 Release - Python 3.11.2". Under this, there is a section for "Stable Releases". The first stable release is "Python 3.10.10 - Feb. 8, 2023". Below this, there is a note: "Note that Python 3.10.10 cannot be used on Windows 7 or earlier." This is followed by a list of download links: "Download Windows embeddable package (32-bit)", "Download Windows embeddable package (64-bit)", "Download Windows help file", "Download Windows installer (32-bit)", and "Download Windows installer (64-bit)". The second stable release is "Python 3.11.2 - Feb. 8, 2023". Below this, there is another note: "Note that Python 3.11.2 cannot be used on Windows 7 or earlier." This is followed by a list of download links: "Download Windows embeddable package (32-bit)", "Download Windows embeddable package (64-bit)", "Download Windows embeddable package (ARM64)", "Download Windows installer (32-bit)", "Download Windows installer (64-bit)", and "Download Windows installer (ARM64)".

python™

About Downloads Documentation

Python >>> Downloads >>> Windows

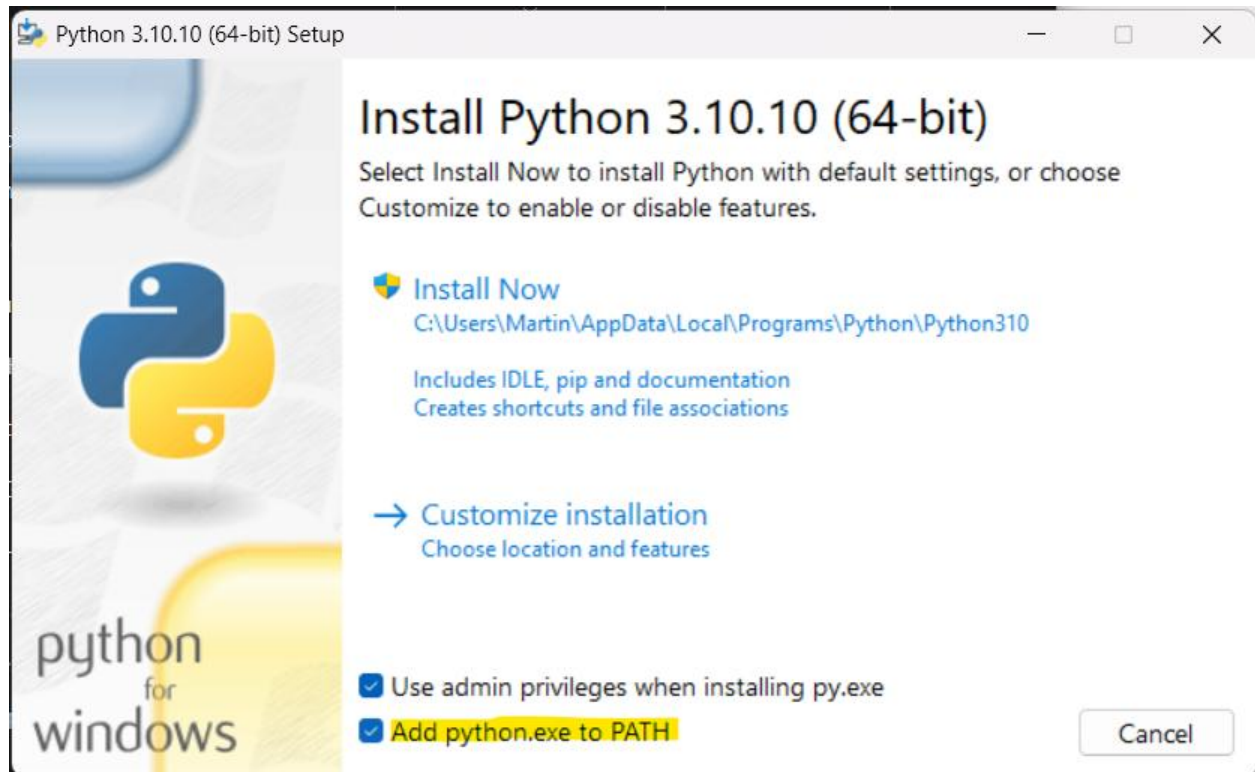
Python Releases for Windows

- [Latest Python 3 Release - Python 3.11.2](#)

Stable Releases

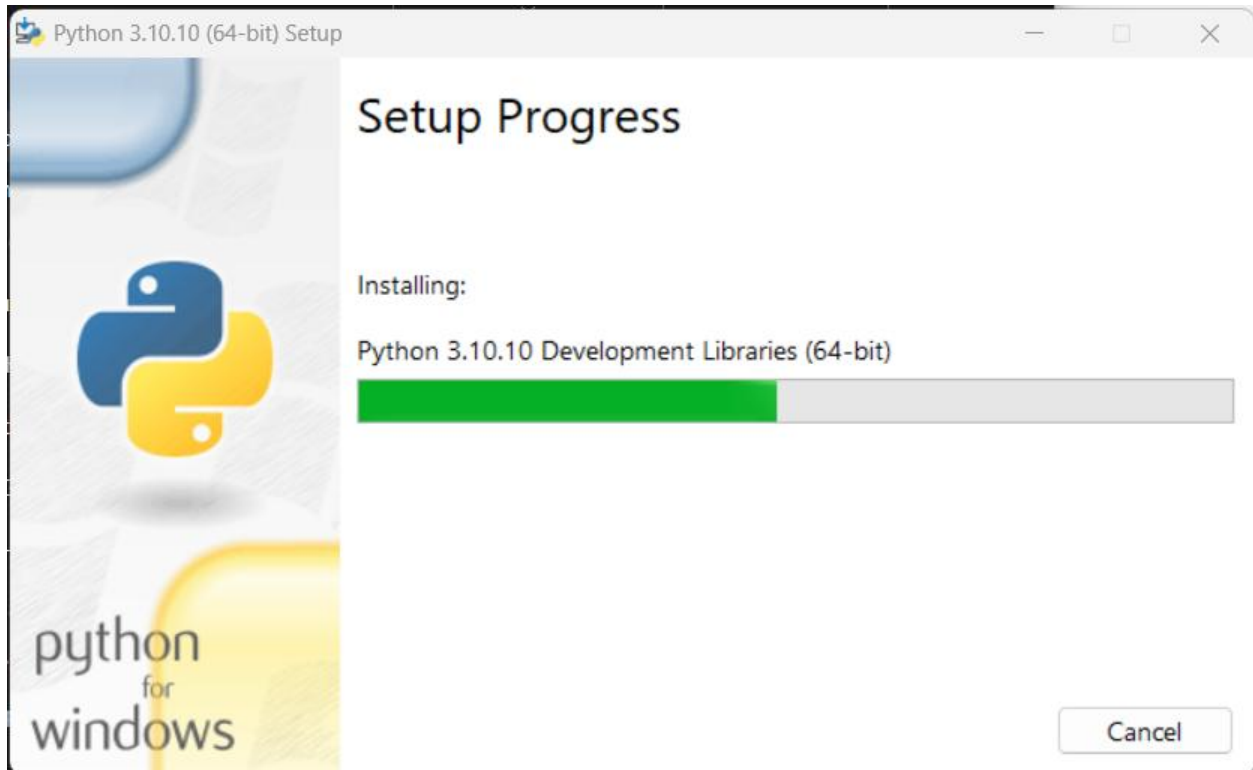
- [Python 3.10.10 - Feb. 8, 2023](#)
Note that Python 3.10.10 cannot be used on Windows 7 or earlier.
 - Download [Windows embeddable package \(32-bit\)](#)
 - Download [Windows embeddable package \(64-bit\)](#)
 - Download [Windows help file](#)
 - Download [Windows installer \(32-bit\)](#)
 - Download [Windows installer \(64-bit\)](#)
- [Python 3.11.2 - Feb. 8, 2023](#)
Note that Python 3.11.2 cannot be used on Windows 7 or earlier.
 - Download [Windows embeddable package \(32-bit\)](#)
 - Download [Windows embeddable package \(64-bit\)](#)
 - Download [Windows embeddable package \(ARM64\)](#)
 - Download [Windows installer \(32-bit\)](#)
 - Download [Windows installer \(64-bit\)](#)
 - Download [Windows installer \(ARM64\)](#)

- You typically want to download the **Windows Installer (64-bit)** for modern hardware. Unless you *know* you need the 32-bit version, try the 64-bit first.
- Once the download is complete, double-click the installer to launch it.



- You'll see the installer welcome screen which describes what the installer will do.
- 📌 Make sure that Add python.exe to PATH is checked in the installer. This makes it easier to use Python from the command prompt -- you just need to enter "python" to start, rather than the full path to the executable.

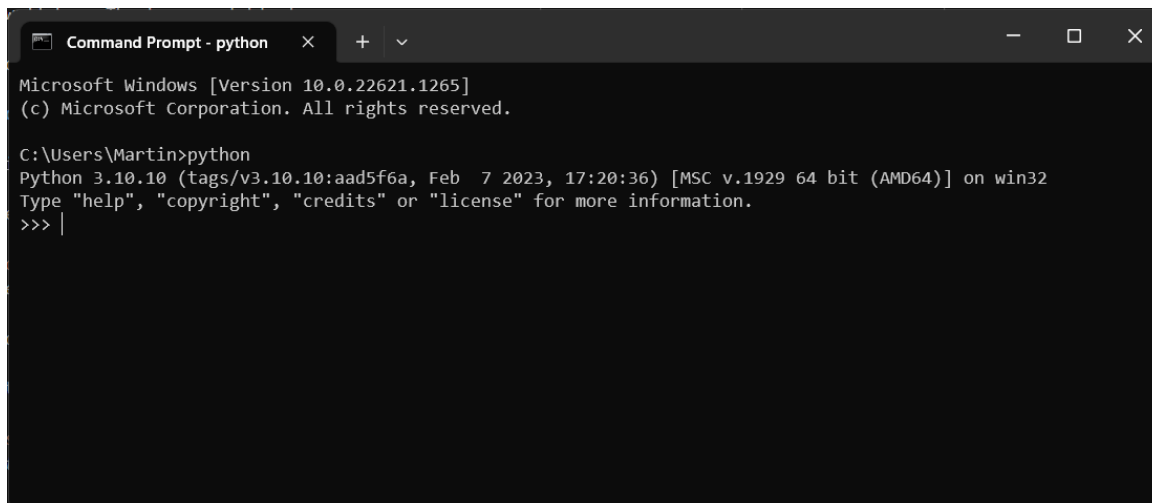
- When you are ready to begin the installation you can click *Install Now*.



- The install will proceed as normal, installing all the required libraries. Once complete you can exit the installer.

Verify Your Python Installation

- Open a command prompt and start Python by entering python. This will start the Python REPL, where you can enter interactive Python code.



- If you see the Python version number and the >>> prompt, your installation was successful.

Troubleshooting Common Installation Issues

If typing python in the command prompt gives you an error such as 'python' is not recognized, the most likely cause is that Python was not added to your system PATH. You can fix this by re-running the installer, selecting **Modify**, and ensuring the "Add python.exe to PATH" option is checked.

Another common issue on Windows 10 and 11 is that typing python opens the Microsoft Store instead of the interpreter. To resolve this, open **Settings > Apps > Advanced app execution aliases** and disable the "App Installer" entries for python.exe and python3.exe.

🍏 How to Install Python on macOS

Method 1: Installing Python Using Homebrew (Recommended)

Homebrew is a popular package manager for macOS that makes installing and managing software incredibly easy. This is now the preferred method for many developers because it simplifies updating Python and installing additional packages. Here's how to do it:

1. First, install Homebrew if you don't have it already. Open a Terminal window and paste the following command:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

2. Once Homebrew is installed, install Python by typing:

```
brew install python@3.10
```

3. After installation completes, verify it worked by typing:

```
python3 --version
```

You should see output indicating that Python 3.10.x is installed.

Method 2: Installing Python Using the Official Installer

If you prefer not to use Homebrew or need a specific Python distribution directly from the source, downloading Python on Mac from the official Python website is also a reliable method. This method gives you the exact version released by the Python Software Foundation without any modifications.

1. First, download an installer package from the Python website. To do that, visit <https://www.python.org/downloads/> on your Mac; it detects your operating system automatically and shows a big button for downloading the latest version of Python installer on your Mac. If it doesn't, click the macOS link and choose the latest Python release.



The screenshot shows the Python.org website. The top navigation bar includes links for Python, PSF, Docs, PyPI, Jobs, and Community. Below this is a search bar and a 'Donate' button. The main content area features a large blue banner with the text 'Download the latest version for macOS' and a prominent yellow button labeled 'Download Python 3.10.1'. To the right of the button is an illustration of two parachutes carrying boxes. Below the button, there are links for other operating systems (Windows, Linux/UNIX, macOS, Other), a link for prereleases, and a link for Docker images. At the bottom of the banner, it says 'Looking for Python 2.7? See below for specific releases'.

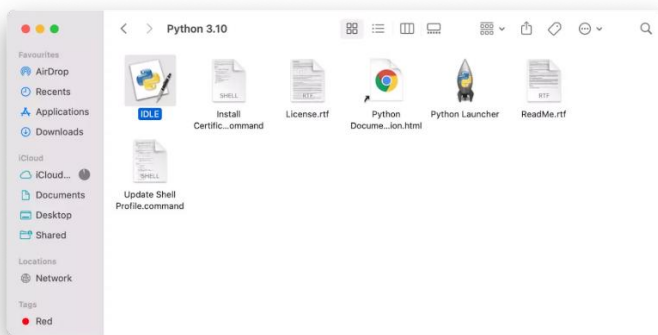
Active Python Releases
For more information visit the [Python Developer's Guide](#).

Python version	Maintenance status	First released	End of support	Release schedule
3.10	bugfix	2021-10-04	2026-10	PEP 619
3.9	bugfix	2020-10-05	2025-10	PEP 596

2. Once the download is complete, double-click the package to start installing Python. The installer will walk you through a wizard to complete the installation, and in most cases, the default settings work well, so install it like the other applications on macOS. You may also have to enter your Mac password to let it know that you agree with installing Python.



3. When the installation completes, it will open up the Python folder.



4. You can verify that Python installed correctly by opening IDLE, which is the integrated development environment that comes with Python. If everything works correctly, IDLE shows the Python shell as follows:



 **NOTE:** The advantage of this installation method is that you can easily update an outdated Python install by downloading the latest Python installer. The new version of Python is available on your Mac once the installation is complete.

Install Jupyter Notebook

Jupyter Notebook is a tool that comes as a Python package. We'll install it using a tool called **pip**, which is included with Python.

1. **Open Command Prompt or Terminal:**

- On Windows, search for **Command Prompt** in the Start menu.
- On macOS/Linux, open the **Terminal** application.

2. **Install Jupyter Notebook:**

- Type the following command into Command Prompt or Terminal and press Enter:

```
pip install notebook
```

- Wait for a few moments while Jupyter Notebook is downloaded and installed.

Now that Jupyter is installed, let's learn how to open it.

Launch Jupyter Notebook

We're now ready to launch Jupyter Notebook and open its interface in your web browser.

1. **Navigate to Your Folder:**

- In Command Prompt or Terminal, type **cd** followed by the path to the folder where you want to save your work. For example:

```
cd Documents
```

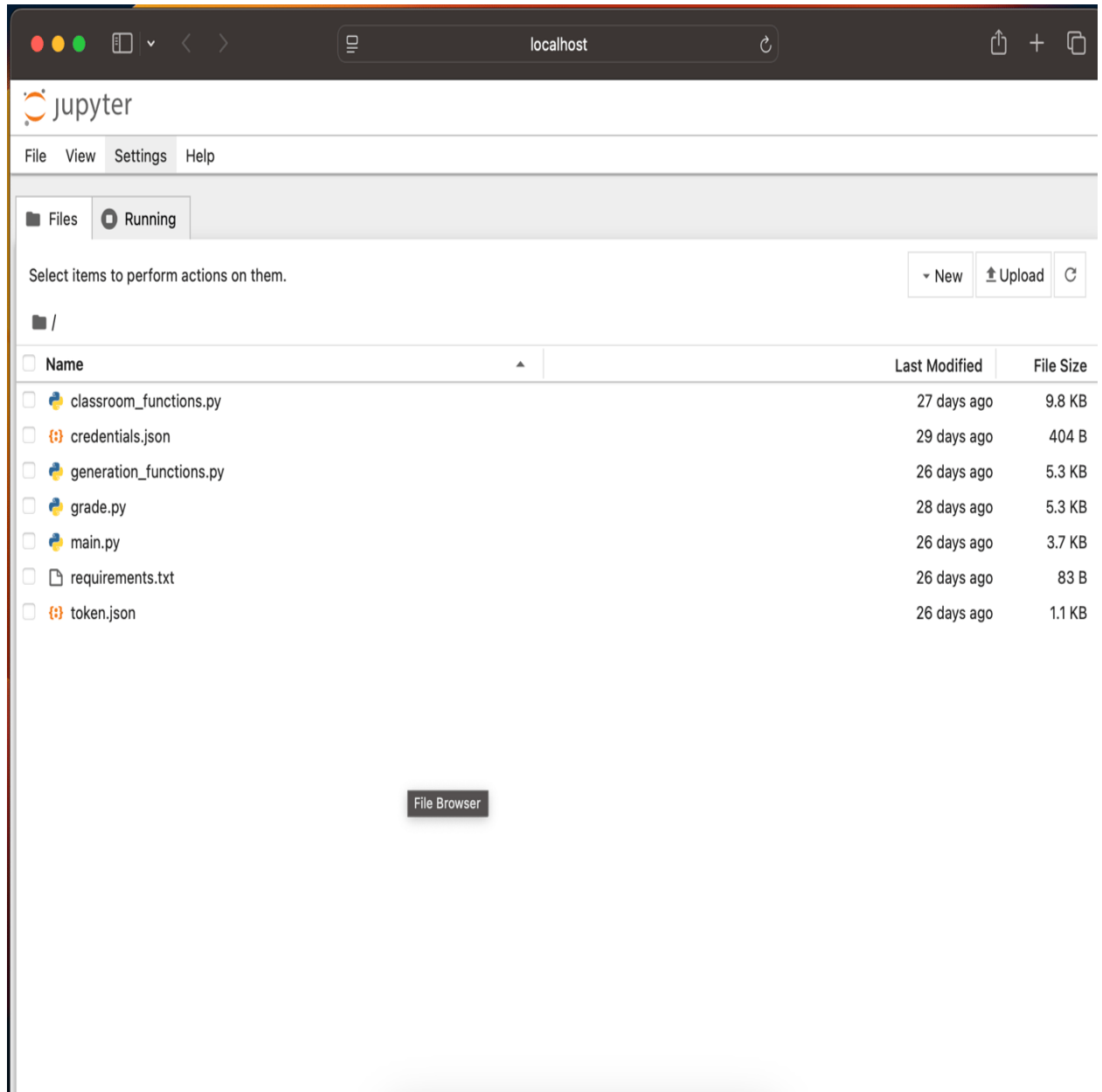
This step tells Jupyter where to save your files.

Start Jupyter Notebook:

- Type this command in the terminal/command prompt and press Enter:

```
jupyter notebook
```

- Your default web browser will open, showing the Jupyter Notebook interface. This is how mine looks like from the directory I entered the command:



If the browser doesn't open automatically, copy the link shown in Command Prompt or Terminal and paste it into your browser.

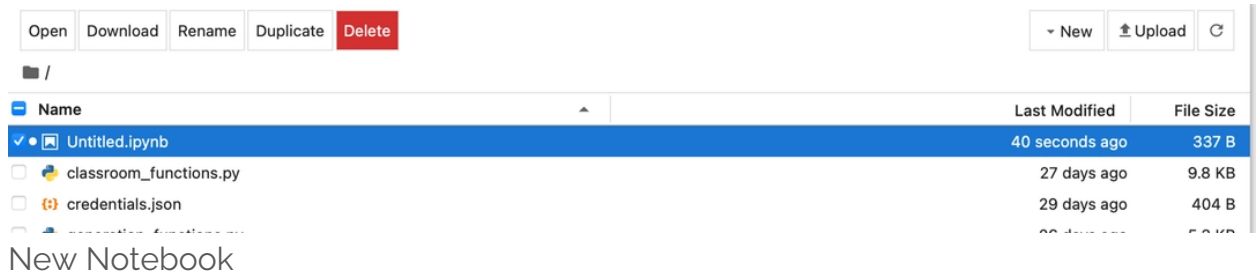
```
llama-classroom — jupyter-notebook — 96x29
[I 2024-11-16 22:57:41.927 LabApp] JupyterLab extension loaded from /opt/anaconda3/envs/llama-cl
assroom/lib/python3.11/site-packages/jupyterlab
[I 2024-11-16 22:57:41.927 LabApp] JupyterLab application directory is /opt/anaconda3/envs/llama
-classroom/share/jupyter/lab
[I 2024-11-16 22:57:41.928 LabApp] Extension Manager is 'pypi'.
[I 2024-11-16 22:57:41.991 ServerApp] jupyterlab | extension was successfully loaded.
[I 2024-11-16 22:57:41.995 ServerApp] notebook | extension was successfully loaded.
[I 2024-11-16 22:57:41.996 ServerApp] Serving notebooks from local directory: /Users/admin/Docum
ents/llama-classroom
[I 2024-11-16 22:57:41.996 ServerApp] Jupyter Server 2.14.2 is running at:
[I 2024-11-16 22:57:41.996 ServerApp] http://localhost:8888/tree?token=b9ec25db1ad28c780f1c92e37
d6d0c07c334a40f6083fc42
[I 2024-11-16 22:57:41.996 ServerApp] http://127.0.0.1:8888/tree?token=b9ec25db1ad28c780f1c9
2e37d6d0c07c334a40f6083fc42
[I 2024-11-16 22:57:41.996 ServerApp] Use Control-C to stop this server and shut down all kernel
s (twice to skip confirmation).
[C 2024-11-16 22:57:42.011 ServerApp]

To access the server, open this file in a browser:
file:///Users/admin/Library/Jupyter/runtime/jpserver-15326-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=b9ec25db1ad28c780f1c92e37d6d0c07c334a40f6083fc42
http://127.0.0.1:8888/tree?token=b9ec25db1ad28c780f1c92e37d6d0c07c334a40f6083fc42
[I 2024-11-16 22:57:42.496 ServerApp] Skipped non-installed server(s): bash-language-server, doc
kerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, julia-la
nguage-server, pyright, python-language-server, python-lsp-server, r-languageserver, sql-languag
e-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver
-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yamll-language-server
```

Create Your First Notebook

Now that Jupyter is running, let's create your very first notebook.

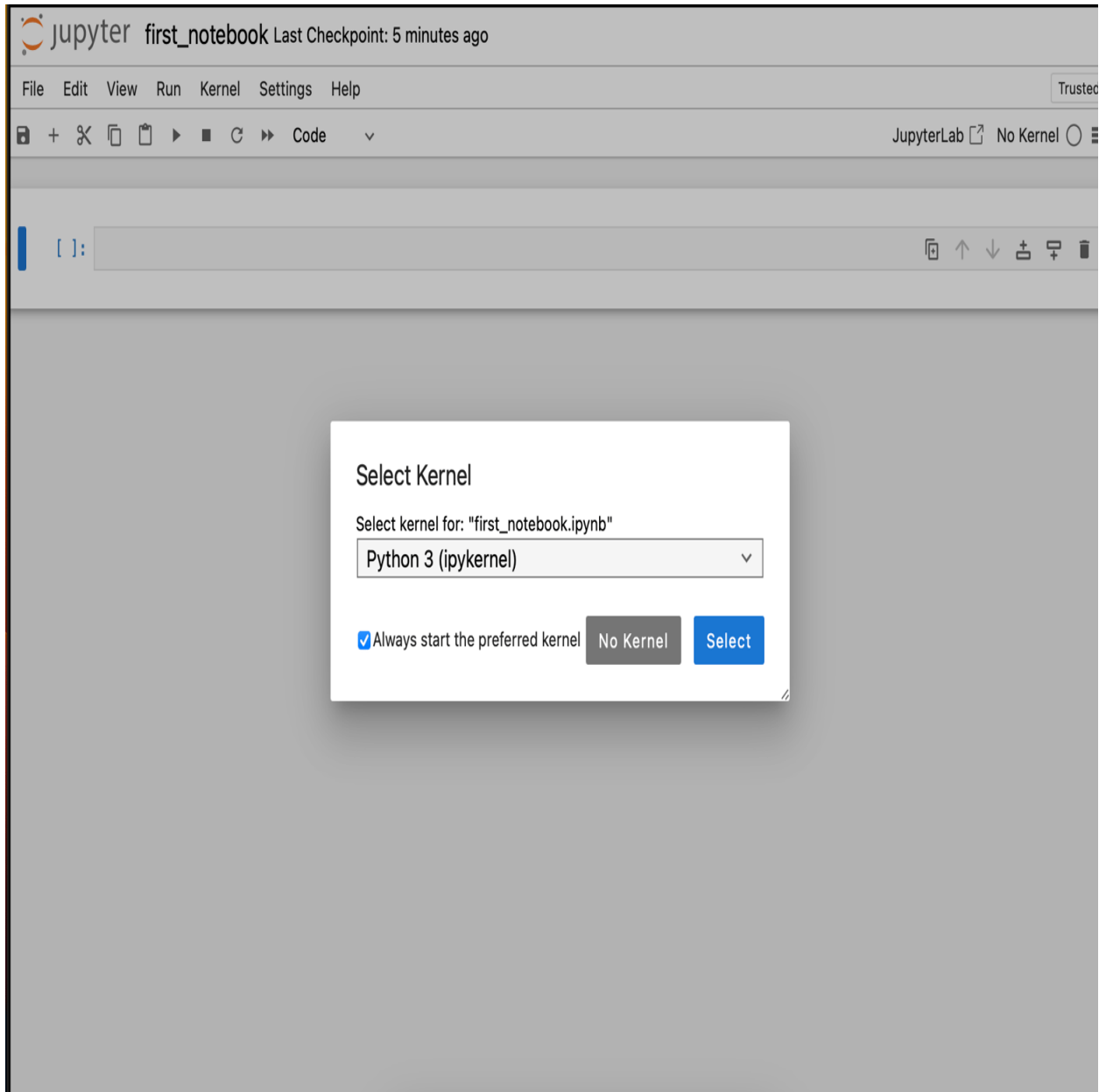
1. In the Jupyter interface, click the **"New"** button on the right side of the screen.
2. From the dropdown menu, select **"Python 3"**.
3. A new notebook will appear in the file list with the default name **Untitled.ipynb**.



4. To rename the file, **right-click** on **Untitled.ipynb** and select **Rename** from the menu.

5. Enter a new name for the file and ensure it ends with **.ipynb** (e.g., **MyFirstNotebook.ipynb**).
6. Double-click the renamed file to open it. You'll be redirected to a new page where you can interact with the kernel and start coding in your notebook.

When the new tab opens, you might see a pop-up box titled **"Select Kernel"** (like the one in the image). If this happens, simply leave **Python 3** as the selected kernel and click **Select**. This ensures your notebook is ready to run Python code.



Select kernel

Let's try running some Python code!

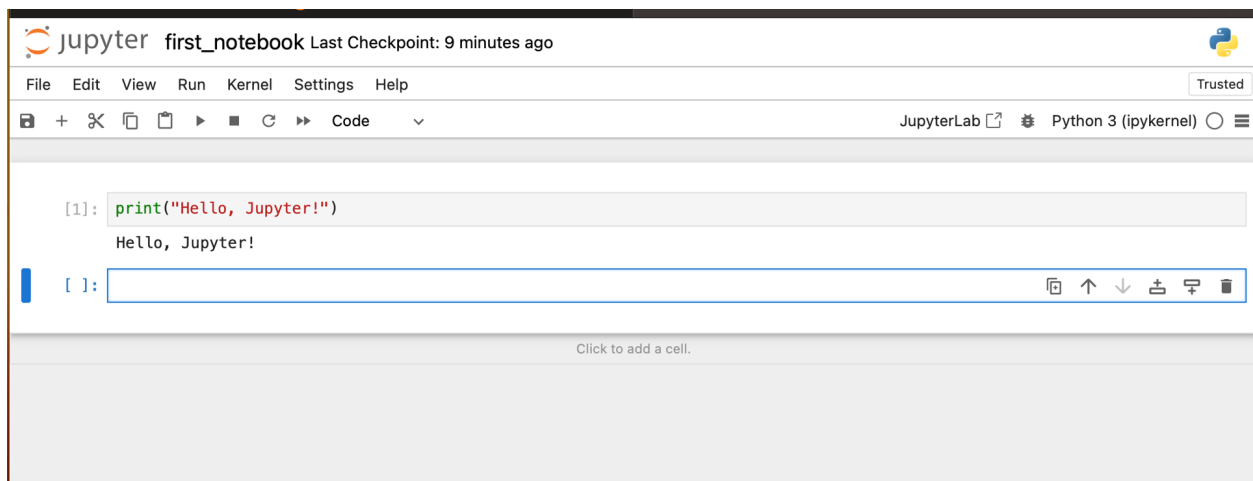
Write and Run Your First Python Code

Jupyter Notebooks are interactive, meaning you can write and execute code one piece at a time.

1. In the empty box (called a “cell”), type the following Python code:

```
print("Hello, Jupyter!")
```

2. Press Shift + Enter on your keyboard to run the code.
3. You'll see the output below the cell—“Hello, Jupyter!”



Save and Close Your Notebook

Once you're done working on your notebook, it's essential to save your progress and close everything properly.

1. **Save Your Work:**
 - To save your notebook, go to the top menu and click on **File > Save Notebook**. Alternatively, you can press **Command + S (macOS)** or **Ctrl + S (Windows)** to save quickly.
2. **Close the Notebook:**
 - To close the notebook, go to **File > Close and Shut Down Notebook**. This ensures the notebook stops running in the background.

3. Stop Jupyter Notebook:

- After closing the notebook, go back to your Command Prompt or Terminal.
- Press **Ctrl + C** and type **Y** when prompted to confirm shutting down the Jupyter Notebook server.

```
llama-classroom — zsh — 96x29
[W 2024-11-16 23:04:27.335 ServerApp] delete /first_notebook.ipynb
[I 2024-11-16 23:05:05.377 ServerApp] Creating new notebook in
[I 2024-11-16 23:05:05.968 ServerApp] Saving file at /Untitled.ipynb
[I 2024-11-16 23:05:05.994 ServerApp] Connecting to kernel 24b19410-a316-48f3-bf77-b2d387cdb52e.
[W 2024-11-16 23:05:06.029 ServerApp] Got events for closed stream <zmq.eventloop.zmqstream.ZMQS
tream object at 0x11357a110>
[W 2024-11-16 23:05:06.031 ServerApp] Got events for closed stream <zmq.eventloop.zmqstream.ZMQS
tream object at 0x113ee10d0>
[W 2024-11-16 23:05:06.031 ServerApp] Got events for closed stream <zmq.eventloop.zmqstream.ZMQS
tream object at 0x113ee10d0>
[I 2024-11-16 23:06:00.596 ServerApp] Connecting to kernel 24b19410-a316-48f3-bf77-b2d387cdb52e.
[I 2024-11-16 23:13:32.267 ServerApp] Kernel started: d3dc2607-930c-4466-a3be-a201cea70c31
[I 2024-11-16 23:13:34.290 ServerApp] Connecting to kernel d3dc2607-930c-4466-a3be-a201cea70c31.
[I 2024-11-16 23:14:02.368 ServerApp] Saving file at /first_notebook.ipynb
[I 2024-11-16 23:16:02.534 ServerApp] Saving file at /first_notebook.ipynb
[I 2024-11-16 23:16:44.047 ServerApp] Saving file at /first_notebook.ipynb
[I 2024-11-16 23:18:41.958 ServerApp] Kernel shutdown: d3dc2607-930c-4466-a3be-a201cea70c31
^C[I 2024-11-16 23:19:17.473 ServerApp] interrupted
[I 2024-11-16 23:19:17.489 ServerApp] Serving notebooks from local directory: /Users/admin/Docum
ents/llama-classroom
1 active kernel
Jupyter Server 2.14.2 is running at:
http://localhost:8888/tree?token=b9ec25db1ad28c780f1c92e37d6d0c07c334a40f6083fc42
http://127.0.0.1:8888/tree?token=b9ec25db1ad28c780f1c92e37d6d0c07c334a40f6083fc42
Shut down this Jupyter server (y/[n])? y
[C 2024-11-16 23:19:19.487 ServerApp] Shutdown confirmed
[I 2024-11-16 23:19:19.511 ServerApp] Shutting down 5 extensions
[I 2024-11-16 23:19:19.514 ServerApp] Shutting down 1 kernel
[I 2024-11-16 23:19:19.521 ServerApp] Kernel shutdown: 24b19410-a316-48f3-bf77-b2d387cdb52e
```

Python Libraries for Data Engineering

Data Processing & Analysis

- **Pandas** – Data cleaning, transformation, aggregation, and analysis.
- **NumPy** – Fast numerical computing and array operations.
- **Polars** – High-performance DataFrame library, often faster than Pandas.
- **Dask** – Parallel and distributed data processing for large datasets.

Database Connectivity

- **SQLAlchemy** – Database ORM and SQL toolkit.
 - **psycopg2** – PostgreSQL database connector.
 - **PyMySQL** – MySQL database connector.
 - **sqlite3** – Built-in SQLite database library.
 - **pyodbc** – Connects to SQL Server, Oracle, and other ODBC-supported databases.
-

ETL & Data Pipelines

- **Apache Airflow** – Workflow orchestration and scheduling.
 - **Prefect** – Modern workflow automation platform.
 - **Luigi** – Batch processing and dependency management.
 - **Bonobo** – Lightweight ETL framework.
-

Cloud Data Engineering

- **boto3** – AWS SDK for Python.
 - **google-cloud-storage** – Google Cloud Storage integration.
 - **google-cloud-bigquery** – BigQuery operations.
 - **azure-storage-blob** – Azure Blob Storage access.
-

File Handling

- **openpyxl** – Read and write Excel files (.xlsx).
 - **xlsxwriter** – Create Excel reports.
 - **pyarrow** – Apache Arrow and Parquet support.
 - **fastparquet** – Read and write Parquet files.
 - **json** – Built-in JSON processing.
 - **csv** – Built-in CSV processing.
-

APIs & Data Extraction

- **requests** – HTTP requests and REST APIs.
 - **httpx** – Modern asynchronous HTTP client.
 - **aiohttp** – Async API calls.
 - **BeautifulSoup (bs4)** – Web scraping.
 - **Scrapy** – Large-scale web scraping framework.
-

Big Data Processing

- **PySpark** – Apache Spark for distributed processing.
 - **PyFlink** – Apache Flink integration.
 - **Ray** – Distributed computing framework.
-

Streaming Data

- **kafka-python** – Apache Kafka client.
 - **confluent-kafka** – High-performance Kafka client.
 - **faust-streaming** – Stream processing with Kafka.
-

Data Quality & Validation

- **Great Expectations** – Data validation and quality checks.
 - **Pandera** – Schema validation for Pandas DataFrames.
 - **pydantic** – Data validation using Python type hints.
-

Monitoring & Logging

- **logging** – Built-in Python logging framework.
- **loguru** – Simplified logging.
- **prometheus-client** – Metrics collection for monitoring.

Useful for Modern Data Platforms

- **dbt-core** – Data transformation in analytics engineering.
 - **DuckDB** – Fast analytical SQL database.
 - **Delta Lake** – Data lakehouse support.
 - **Pylceberg** – Apache Iceberg table format.
-

Top 10 Libraries Every Beginner Data Engineer Should Learn

1. Pandas
2. NumPy
3. requests
4. SQLAlchemy
5. psycopg2
6. PySpark
7. Apache Airflow
8. openpyxl
9. boto3
10. Great Expectations

A typical Data Engineer's toolkit today often includes:

Python

├── Pandas
├── NumPy
├── Requests
├── SQLAlchemy
├── Psycopg2
├── PySpark
├── Airflow
├── PyArrow

└── Boto3
└── Great Expectations

These libraries cover the core Data Engineering workflow: **Extract** → **Transform** → **Load (ETL)** → **Store** → **Validate** → **Monitor** → **Orchestrate**. 🚀

Here are the pip install commands for the most common Data Engineering libraries:

Data Processing

pip install pandas

pip install numpy

pip install polars

pip install dask

Database Connectivity

pip install sqlalchemy

pip install psycopg2-binary

pip install pymysql

pip install pyodbc

ETL & Workflow Orchestration

pip install apache-airflow

pip install prefect

pip install luigi

Cloud

pip install boto3

pip install google-cloud-storage

pip install google-cloud-bigquery

```
pip install azure-storage-blob
```

File Handling

```
pip install openpyxl
```

```
pip install xlswriter
```

```
pip install pyarrow
```

```
pip install fastparquet
```

APIs & Data Extraction

```
pip install requests
```

```
pip install httpx
```

```
pip install aiohttp
```

```
pip install beautifulsoup4
```

```
pip install scrapy
```

Big Data

```
pip install pyspark
```

```
pip install ray
```

Streaming

```
pip install kafka-python
```

```
pip install confluent-kafka
```

Data Quality

```
pip install great-expectations
```

```
pip install pandera
```

```
pip install pydantic
```

Monitoring & Logging

pip install loguru

pip install prometheus-client

Modern Data Platforms

pip install dbt-core

pip install duckdb

pip install pyiceberg

Install Everything at Once

pip install pandas numpy polars dask sqlalchemy psycpg2-binary pymysql
pyodbc requests httpx aiohttp openpyxl xlsxwriter pyarrow fastparquet pyspark
kafka-python confluent-kafka great-expectations pandera pydantic loguru
duckdb boto3

Verify Installation

pip list

or

pip show pandas

Recommended for a Beginner Data Engineering Course

pip install pandas numpy requests sqlalchemy psycpg2-binary openpyxl
pyarrow pyspark jupyter

This set is enough to learn:

- CSV processing
- JSON processing
- API data extraction
- PostgreSQL connectivity
- ETL pipelines
- Excel files

- Parquet files
- Big Data with Spark
- Jupyter Notebook development 🚀